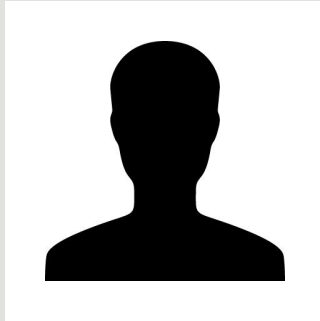


Quantum Advancements

Quantum advancements compared between
manufacturers i.e IBM & Google

Presenters



Ibrahim
Engineering
mi2171@nyu.edu

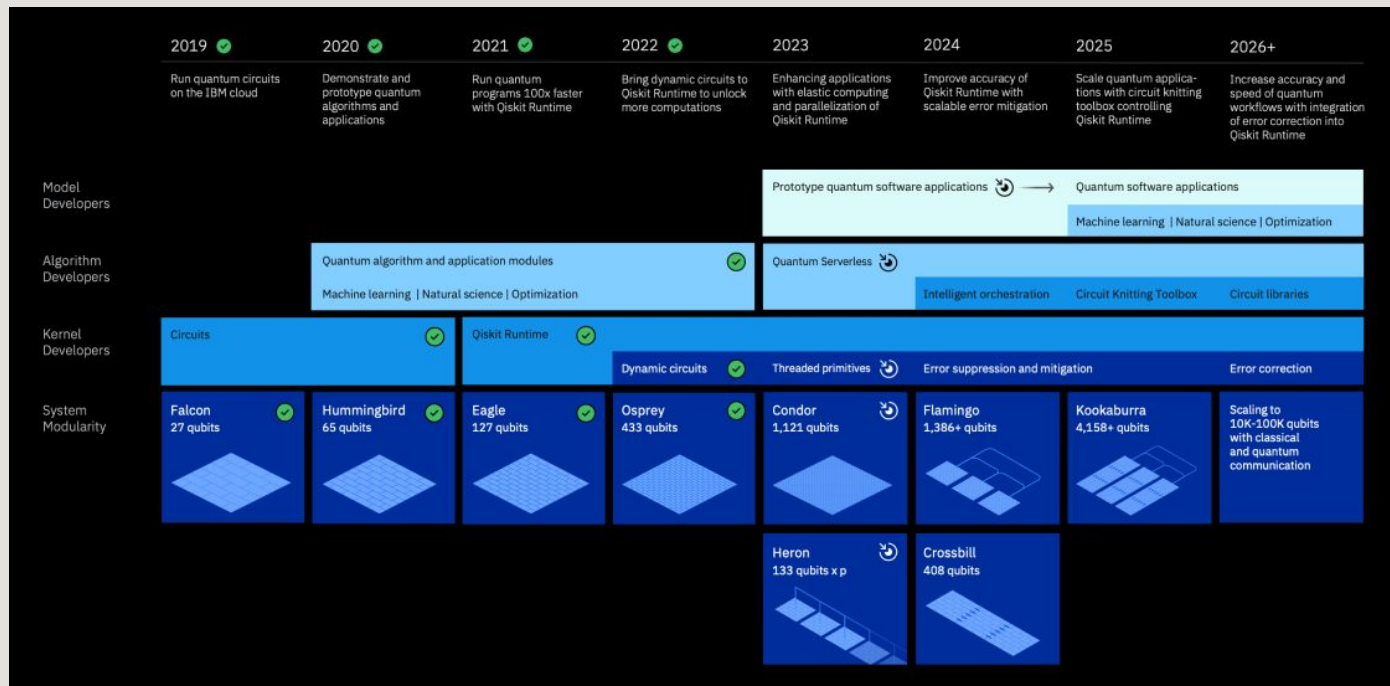
IBM vs Google

IBM is currently leading the quantum market with their 2022 Roadmap, in which they have achieved a quantum chip that consists of 433 qubits named Osprey, collectively named IBM Quantum System One. They also provide an online Quantum lab that users can use to find out and learn about their algorithms.

However, rivals Google are also in the chase, with a much weaker processor still in the works. They have a 54 qubit processor, according to their website and also provide an online IDE in which users can advance and develop their algorithms.

In this presentation, we will focus on IBM's advancements.

IBM development Roadmap



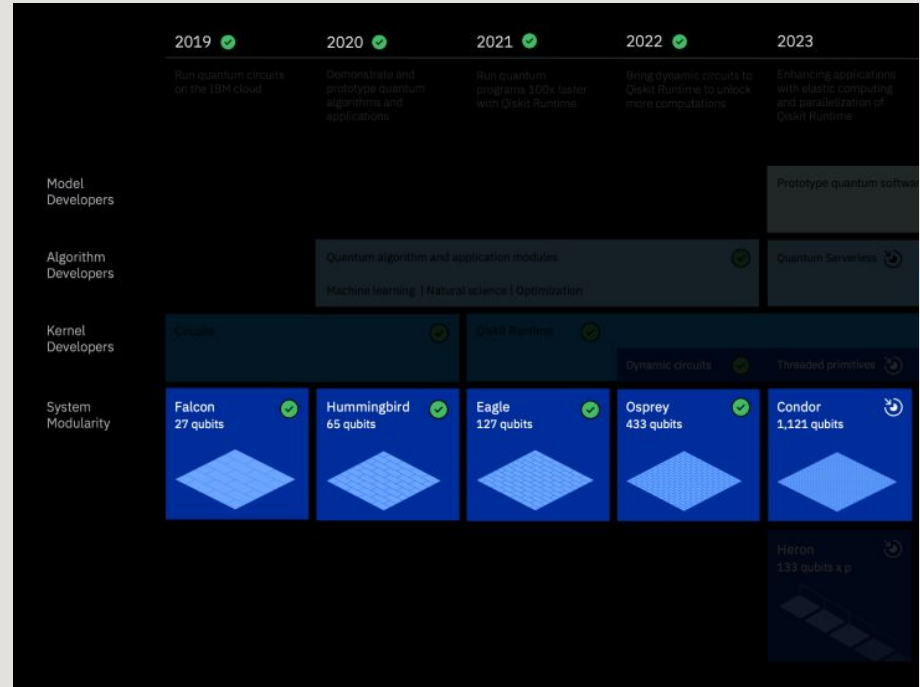
Speed, Scale and Quality

2021

100-qubit barrier broken with 127-qubit Eagle processor, just two years after the debut of the 27-qubit Falcon chip.

2022

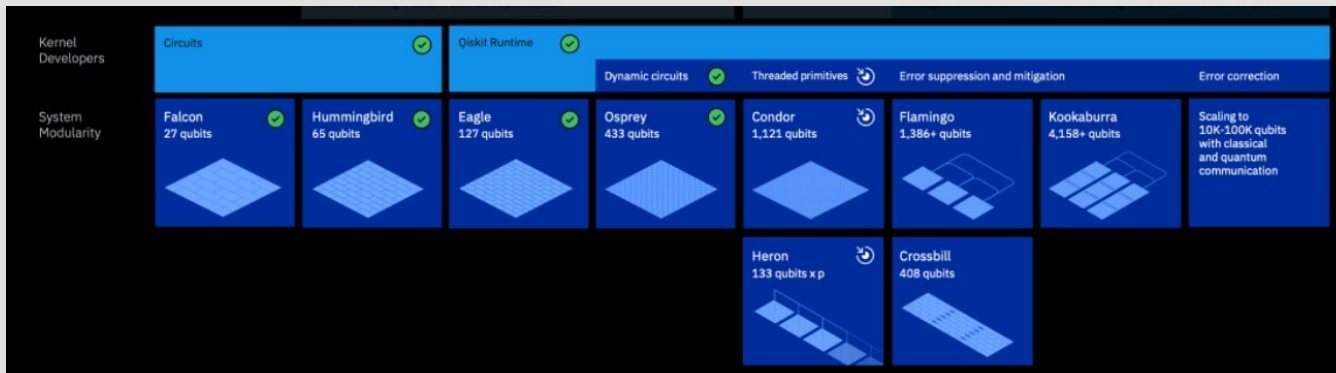
433-qubit Osprey processor and deliver the 1000+ qubit Condor processor the year after.



Error and Noise

2022

Dynamic Circuits. These extend what the hardware can do by reducing circuit depth, allowing for alternative models of circuit construction, and enable fundamental operations at the heart of quantum error correction.



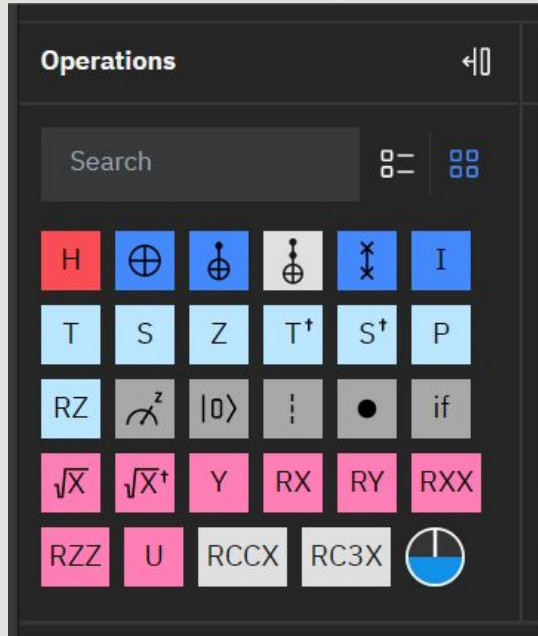
IBM Quantum Composer

The screenshot displays the IBM Quantum Composer web application. The interface is divided into several sections:

- Top Bar:** Includes the title "IBM Quantum Composer", a search icon, a help icon, and a user profile icon. Below this is a menu bar with "My first circuit" (marked as "Saved"), "File", "Edit", and "View". On the right, it shows "Visualizations seed 3290" and a "Setup and run" button.
- Operations Panel:** Located on the left, it features a search bar and a grid of quantum gates. The gates are categorized by color: red (H, T, RZ, $\sqrt{X}$, RZZ), blue (CNOT, S, Z, T[†], S[†], P), green (CNOT, X, Y, RX, RY, RXX), and purple (U, RCCX, RC3X). There are also icons for state preparation and measurement.
- Circuit Editor:** The central area shows a quantum circuit with three qubits: q[0], q[1], and c2. Each qubit line has a circular icon at the end, likely for zooming or deleting the line.
- Probabilities Panel:** Located at the bottom left, it displays a line graph titled "Probabilities" with "Probability (%)" on the y-axis (0 to 100) and "Computational basis states" on the x-axis. The graph is currently empty.
- Q-sphere Panel:** Located at the bottom right, it shows a Bloch sphere visualization. A blue dot is positioned at the top of the sphere, labeled "000". Below the sphere is a circular gauge for "Phase" (0 to $2\pi/2$) and checkboxes for "State" (checked) and "Phase angle".
- Code Editor:** On the far right, there is a code editor showing OpenQASM 2.0 code. The code includes a header, an include statement for "qelib1.inc", and declarations for qubits and a classical register.

```
1 OPENQASM 2.0;  
2 include "qelib1.inc";  
3 qreg q[2];  
4 creg c[2];
```

Quantum Gates

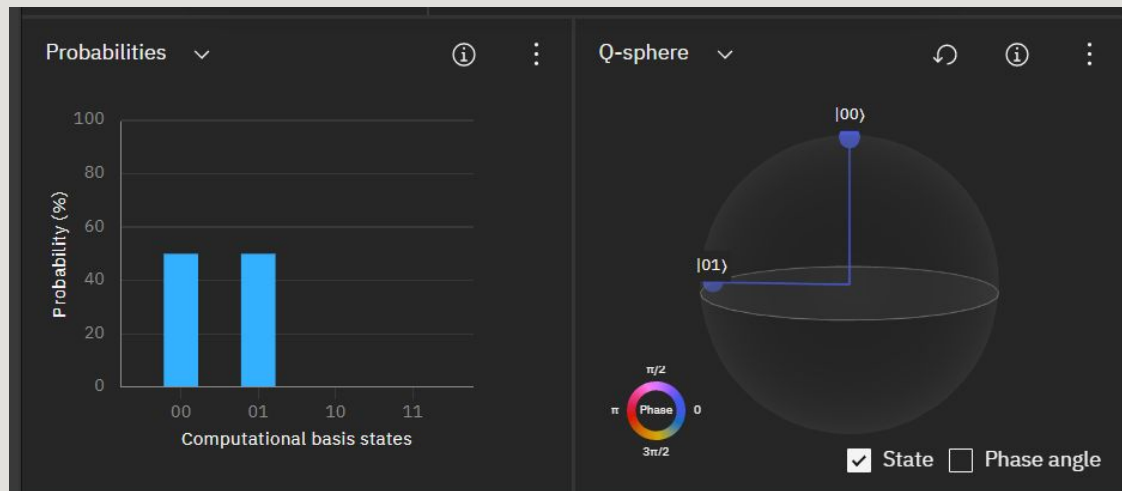


Gates and operations manipulate qubits. They are the building blocks of quantum circuits.

You will drag and drop operation blocks onto the wires on the right to construct your quantum circuit.

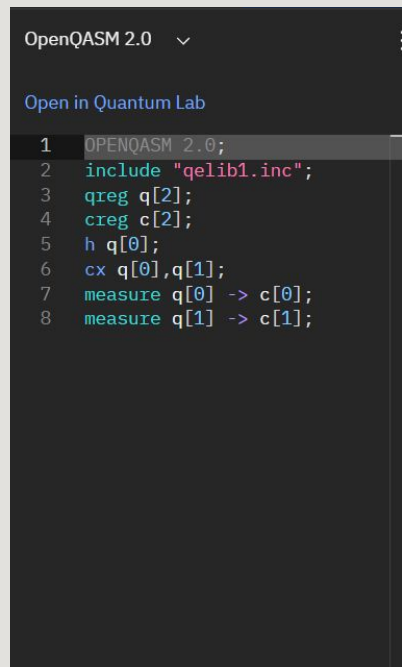
Visual and Quantitative Output

Visualizations use a simulator to show what's happening on a quantum computer before you measure your circuit. They are updated each time you add an operation to a wire.



Code Editor

The code editor updates as you add operations to your circuit. If you update the code, changes will be reflected in the graphical editor.

A screenshot of a code editor window titled "OpenQASM 2.0". At the top, there is a button labeled "Open in Quantum Lab". Below this, the code is displayed with line numbers 1 through 8 on the left. The code is as follows:

```
1 OPENQASM 2.0;  
2 include "qelib1.inc";  
3 qreg q[2];  
4 creg c[2];  
5 h q[0];  
6 cx q[0],q[1];  
7 measure q[0] -> c[0];  
8 measure q[1] -> c[1];
```

References

<https://quantum-computing.ibm.com/composer/files/2a126cbf533fbbf8ce60e84b2616a0043782015916bc85cae16825a7327bb39d>

<https://quantum-computing.ibm.com/>